

```
R1#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#no ip route 40.0.0.0 255.0.0.0 20.0.0.2
R1(config)#exit
R1#s
*Jul 20 08:30:21.852: %SYS-5-CONFIG_I: Configured from console by console
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```

      10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, Ethernet0/0
L       10.0.0.1/32 is directly connected, Ethernet0/0
S      15.0.0.0/8 [1/0] via 20.0.0.2
      20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, Ethernet0/1
L       20.0.0.1/32 is directly connected, Ethernet0/1
S      21.0.0.0/8 [1/0] via 20.0.0.2
S      30.0.0.0/8 [1/0] via 20.0.0.2
S     162.14.0.0/16 [1/0] via 20.0.0.2
S     172.19.0.0/16 [1/0] via 20.0.0.2
S     192.168.168.0/24 [1/0] via 20.0.0.2
```

```
R2(config-if)#no shutdown
R2(config-if)#exit
R2(config)#exit
R2#show
*Jul 20 08:34:16.578: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip int b

```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	20.0.0.2	YES	manual	up	up
Ethernet0/1	30.0.0.1	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R2#
R2#config t
Enter configuration commands, one per line. End with CNTL/Z.
R2(config)#ip route 10.0.0.0 255.0.0.0 20.0.0.1
R2(config)#ip route 172.19.0.0 255.255.0.0 30.0.0.2
R2(config)#ip route 15.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 21.0.0.0 255.0.0.0 30.0.0.2
R2(config)#ip route 192.168.168.0 255.255.255.0 30.0.0.2
R2(config)#ip route 162.14.0.0 255.255.0.0 30.0.0.2
R2(config)#exit
R2#
R2#
*Jul 20 08:38:34.134: %SYS-5-CONFIG_I: Configured from console by console
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

S    10.0.0.0/8 [1/0] via 20.0.0.1
S    15.0.0.0/8 [1/0] via 30.0.0.2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     20.0.0.0/8 is directly connected, Ethernet0/0
L     20.0.0.2/32 is directly connected, Ethernet0/0
S    21.0.0.0/8 [1/0] via 30.0.0.2
     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     30.0.0.0/8 is directly connected, Ethernet0/1
L     30.0.0.1/32 is directly connected, Ethernet0/1
S    162.14.0.0/16 [1/0] via 30.0.0.2
S    172.19.0.0/16 [1/0] via 30.0.0.2
S    192.168.168.0/24 [1/0] via 30.0.0.2
```



```
R3#show
*Jul 20 08:48:21.053: %SYS-5-CONFIG_I: Configured from console by console
```

```
R3#show ip int b
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	30.0.0.2	YES	manual	up	up
Ethernet0/1	172.19.0.1	YES	manual	up	up
Ethernet0/2	21.0.0.1	YES	manual	up	up
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R3#show ip route
```

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 30.0.0.1
S    15.0.0.0/8 [1/0] via 172.19.0.2
S    20.0.0.0/8 [1/0] via 30.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        21.0.0.0/8 is directly connected, Ethernet0/2
L        21.0.0.1/32 is directly connected, Ethernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C        30.0.0.0/8 is directly connected, Ethernet0/0
L        30.0.0.2/32 is directly connected, Ethernet0/0
S    162.14.0.0/16 [1/0] via 21.0.0.2
C    172.0.0.0/8 is directly connected, Ethernet0/1
    172.19.0.0/32 is subnetted, 1 subnets
L        172.19.0.1 is directly connected, Ethernet0/1
S    192.168.168.0/24 [1/0] via 21.0.0.2
```

```
*Jul 20 08:52:56.631: %SYS-5-CONFIG_I: Configured from console by console
```

```
R4#show ip int b
```

Interface	IP-Address	OK?	Method	Status	Protocol
Ethernet0/0	15.0.0.1	YES	manual	up	up
Ethernet0/1	172.19.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	TFTP	down	down
Ethernet0/3	unassigned	YES	TFTP	down	down

```
R4#
```

```
R4#
```

```
R4#
```

```
R4#config t
```

```
Enter configuration commands, one per line. End with CNTL/Z.
```

```
R4(config)#
```

```
*Jul 20 08:53:09.199: %PNP-6-PNP_DISCOVERY_STOPPED: PnP Discovery stopped (Config Wizard)
```

```
R4(config)#ip route 10.0.0.0 255.0.0.0 172.19.0.1
```

```
R4(config)#ip route 20.0.0.0 255.0.0.0 172.19.0.1
```

```
R4(config)#ip route 30.0.0.0 255.0.0.0 172.19.0.1
```

```
R4(config)#ip route 21.0.0.0 255.0.0.0 172.19.0.1
```

```
R4(config)#ip route 192.168.168.0 255.255.255.0 172.19.0.1
```

```
R4(config)#ip route 162.14.0.0 255.255.0.0 172.19.0.1
```

```
R4(config)#exit
```

```
R4#
```

```
*Jul 20 08:57:48.056: %SYS-5-CONFIG_I: Configured from console by console
```

```
R4#show ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2
        i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
        ia - IS-IS inter area, * - candidate default, U - per-user static route
        o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
        a - application route
        + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    10.0.0.0/8 [1/0] via 172.19.0.1
    15.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
```

```
C    15.0.0.0/8 is directly connected, Ethernet0/0
```

```
L    15.0.0.1/32 is directly connected, Ethernet0/0
```

```
S    20.0.0.0/8 [1/0] via 172.19.0.1
```

```
S    21.0.0.0/8 [1/0] via 172.19.0.1
```

```
S    30.0.0.0/8 [1/0] via 172.19.0.1
```

```
S    162.14.0.0/16 [1/0] via 172.19.0.1
```

```
    172.19.0.0/16 is variably subnetted, 2 subnets, 2 masks
```

```
C    172.19.0.0/16 is directly connected, Ethernet0/1
```

```
L    172.19.0.2/32 is directly connected, Ethernet0/1
```

```
S    192.168.168.0/24 [1/0] via 172.19.0.1
```



```
R5#
R5#config t
Enter configuration commands, one per line.  End with CNTL/Z.
R5(config)#ip route 162.14.0.0 255.255.0.0 192.168.168.2
R5(config)#exit
R5#s
*Jul 20 09:19:02.135: %SYS-5-CONFIG_I: Configured from console by console
R5#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 21.0.0.1
S    15.0.0.0/8 [1/0] via 21.0.0.1
S    20.0.0.0/8 [1/0] via 21.0.0.1
    21.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C      21.0.0.0/8 is directly connected, Ethernet0/0
L      21.0.0.2/32 is directly connected, Ethernet0/0
S    30.0.0.0/8 [1/0] via 21.0.0.1
S    162.14.0.0/16 [1/0] via 192.168.168.2
S    172.19.0.0/16 [1/0] via 21.0.0.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.168.0/24 is directly connected, Ethernet0/1
L      192.168.168.1/32 is directly connected, Ethernet0/1
```



R6 con0 is now available

Press RETURN to get started.

R6#show ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    10.0.0.0/8 [1/0] via 192.168.168.1
S    15.0.0.0/8 [1/0] via 192.168.168.1
S    20.0.0.0/8 [1/0] via 192.168.168.1
S    21.0.0.0/8 [1/0] via 192.168.168.1
S    30.0.0.0/8 [1/0] via 192.168.168.1
    162.14.0.0/16 is variably subnetted, 2 subnets, 2 masks
C      162.14.0.0/16 is directly connected, Ethernet0/0
L      162.14.0.1/32 is directly connected, Ethernet0/0
S    172.19.0.0/16 [1/0] via 192.168.168.1
    192.168.168.0/24 is variably subnetted, 2 subnets, 2 masks
C      192.168.168.0/24 is directly connected, Ethernet0/1
L      192.168.168.2/32 is directly connected, Ethernet0/1
```

R6#

R6#

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=3ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 2ms, Maximum = 3ms, Average = 2ms

```
C:\Documents and Settings\user>ping 162.14.0.10
```

```
Pinging 162.14.0.10 with 32 bytes of data:
```

```
Reply from 162.14.0.10: bytes=32 time=4ms TTL=123
```

```
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
```

```
Reply from 162.14.0.10: bytes=32 time=3ms TTL=123
```

```
Reply from 162.14.0.10: bytes=32 time=2ms TTL=123
```

```
Ping statistics for 162.14.0.10:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 2ms, Maximum = 4ms, Average = 2ms
```




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Cisco ASDM-I...

Old Java ASDM

PuTTY

```
C:\WINDOWS\system32\cmd.exe
Reply from 10.0.0.10: bytes=32 time=5ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 2ms

C:\Documents and Settings\user>ping 162.14.0.10

Pinging 162.14.0.10 with 32 bytes of data:

Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>
```



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ASDM-I...

Old Java ASDM

```
C:\WINDOWS\system32\cmd.exe
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124
Reply from 162.14.0.10: bytes=32 time=3ms TTL=124
Reply from 162.14.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 162.14.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 3ms, Average = 2ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=5ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124
Reply from 10.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 2ms

C:\Documents and Settings\user>
```



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Old Java ASDM

PuTTY

```
C:\WINDOWS\system32\cmd.exe
Reply from 30.0.0.2: bytes=32 time=2ms TTL=253
Reply from 30.0.0.2: bytes=32 time=1ms TTL=253
Reply from 30.0.0.2: bytes=32 time=2ms TTL=253
Reply from 30.0.0.2: bytes=32 time=1ms TTL=253

Ping statistics for 30.0.0.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>ping 10.0.0.10

Pinging 10.0.0.10 with 32 bytes of data:

Reply from 10.0.0.10: bytes=32 time=5ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Documents and Settings\user>
```




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ExplorerCisco
ASDM-I...

Old Java ASDM



PuTTY

```
C:\WINDOWS\system32\cmd.exe
Reply from 10.0.0.10: bytes=32 time=5ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123
Reply from 10.0.0.10: bytes=32 time=2ms TTL=123
Reply from 10.0.0.10: bytes=32 time=3ms TTL=123

Ping statistics for 10.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\Documents and Settings\user>ping 15.0.0.10

Pinging 15.0.0.10 with 32 bytes of data:

Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124
Reply from 15.0.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 15.0.0.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 2ms, Maximum = 2ms, Average = 2ms

C:\Documents and Settings\user>
```